

The Grader: measuring a classifier for Tunisian Instagram comments

CS496 AI Engineering, Project 1 (Dr Abdeldjalil Labed), MedTech, Fall 2026. Team members and contributions: see README.md.

1. The task, and why exact match is not enough

Task. A model reads one comment from a real Instagram account (English, French, Arabic script, and Tunisian Darija written with Latin letters and numbers, often with emojis) and answers with a label, *Positive*, *Humour* or *Hate Speech*, and a one-sentence reason. One model call per item (openai/gpt-oss-20b on Groq, temperature 0), no agents and no retrieval. The instructor approved the task.

Why exact match is not enough. The label can be compared with the gold label, and we do that. But (a) the gold label itself is not always certain: two careful labellers disagreed on 10% of the comments, mostly on the border between a joke and an insult; and (b) a right label with a wrong reason is not a good answer, and a reason has no single correct text. So the project measures the label by exact match, and measures the reason with an LLM judge that we test against human grades.

2. The golden set

Data. 200 comments from the account's reels, split into 140 dev (to improve prompts) and 60 test (used once, for the final score). Final labels: Positive 84, Humour 58, Hate Speech 58. No item was dropped. The comments contain Darija, so a labelling guide (data/labelling_guide.md) defines the three labels with examples, a Darija glossary and decision rules.

Labelling. Two labellers labelled every comment alone, without seeing each other's label. A third person chose the final label for the disagreements; in all 20 cases it equals one of the two original labels.

Check	Result
Labeller agreement	180 of 200 (90%) , Cohen's kappa 0.85 (kappa corrects for chance agreement)
Items where they differed	20 of 200 (10%): Humour vs Hate Speech 11, Positive vs Humour 7, Positive vs Hate Speech 2
Items dropped	0

Known limits. Only three labels by choice: there is no Neutral and no Criticism label. Neutral comments land in Positive, and hostile or mocking criticism of a person or group counts as Hate Speech. A three-label scheme therefore merges hate and hostile criticism.

3. The harness

One command runs a prompt on a split and writes **one row per item** (input, gold label, predicted label, reason, correct, error, latency, tokens) to results/<prompt>_<split>.jsonl:

```
python harness/runner.py --prompt prompts/classifier_v3.txt --split dev --delay 8
```

- **Structured output.** The model must answer in a fixed JSON schema (label from a fixed list, plus reason), so scoring never parses free text.
- **Every item gets a row.** If the API fails after retries, the row stores the error and the item counts as wrong. Re-running continues where it stopped and retries only failures. Three comments (POOL0574, POOL0563, POOL0483) always return an empty model answer; they stay in the counts.
- **Scorer.** harness/scorer.py prints counts, per-label recall and precision, the confusion matrix, latency, tokens and cost, and saves metrics_*.json. Prompts live in prompts/ under git, and CHANGELOG.md links every change to the score before and after.
- **Dashboard.** harness/build_dashboard.py turns the metrics into one results page.

4. The judge and its reliability

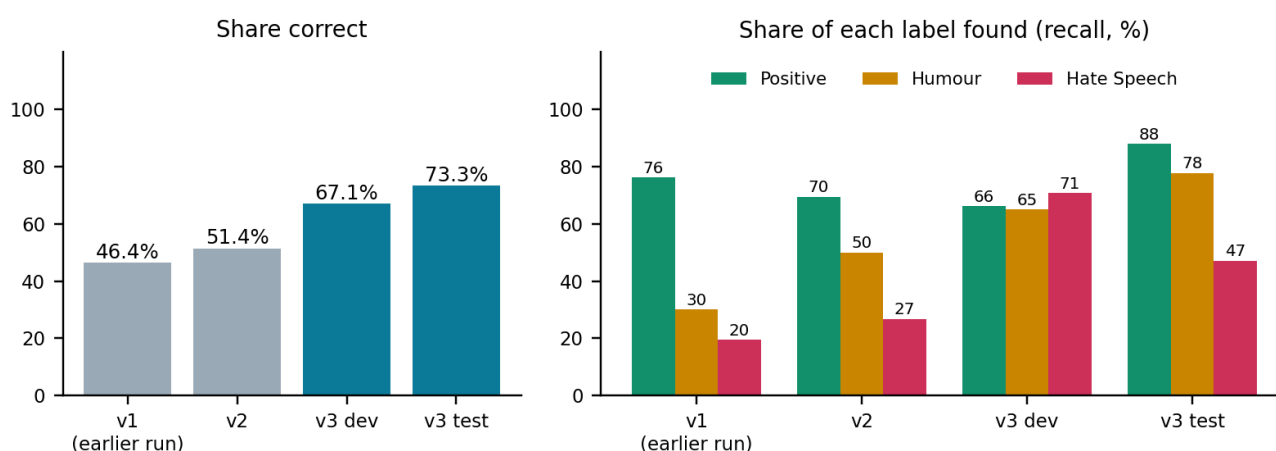
Design. judge_v1 reads the comment and the robot's label and reason, and returns a grade (*good*, *partly*, *bad*) and a short reason in a fixed JSON schema, using the rules of the labelling guide. It does **not** see the gold label. Good = right label and a reason that points at what shows it; partly = right label, vague reason; bad = wrong label, invented context, or an "unclear" reason followed by a harmless label.

Human grades. We drew 40 robot answers (v3 on dev; 20 with a wrong label, 20 with a right label, shuffled). Two people graded each alone, and a third person set the final grade where they differed. By rule, a wrong label is *bad*; humans graded the reasoning on the 20 right-label rows. The two graders agreed on 35 of 40 (87.5%), and on 15 of 20 (75%) of the right-label rows.

Check	Result
Agreement with the final human grade	18 of 39 (46.2%), kappa 0.145
Same, only rows where the robot's label is right	13 of 20 (65.0%), kappa 0.136
Verbosity bias. Padding each reason with a filler sentence (13.9 to 56.9 words)	grade went up for 0 of 39 , stayed the same for 34, went down for 5
Position bias. Same two answers shown in both orders (38 pairs, exactly one has the right label)	verdict changed for 6 of 38 (15.8%) ; the first-shown answer was picked in 34 of 76 calls (45%)
Pairwise: picks the answer with the right label	49 of 76 calls (64.5%), chance is 50%

Where the judge fails. It is too lenient: it said *good* for 32 of 39 answers and caught only 5 of the 21 answers the humans graded *bad*. Because it cannot see the gold label, a wrong label with a fluent reason convinces it. It never used *partly* (humans used it 5 times). The padding test has a limit: 32 of 39 answers already had the top grade, so only 7 could go up. Position changes are not a clear preference for either position, but with 38 pairs the true rate could be roughly 7% to 31%. One item (POOL0563) always returned an empty answer and was not graded. With 40 items, all judge numbers are uncertain.

5. The system: score history



Prompt	Correct	Positive	Humour	Hate Speech	Cost / 1k
v1 (earlier run, prompt text lost)	65/140 (46.4%)	45/59	12/40	8/41	\$0.09
v2	72/140 (51.4%)	41/59	20/40	11/41	\$0.18
v3 on dev	94/140 (67.1%)	39/59	26/40	29/41	\$0.23
v3 on test (run once)	44/60 (73.3%)	22/25	14/18	8/17	\$0.19

What changed. v2 added detailed label guidance. In v2, 21 of 41 Hate Speech comments were called Humour, and most were Darija in Latin letters that the model did not understand. v3 added a short Darija glossary (vulgar words, religious insults, "they are lying", "rich people"), an explicit Hate Speech definition, and rules: Positive is not the default, and an unknown word must not be read as harmless. Dev score 51.4% to 67.1%; Hate Speech found 11/41 to 29/41.

Honest caveat. The glossary was written after reading dev errors, so the dev score is optimistic. The test set shows it: Hate Speech found fell from 70.7% (dev) to **47.1% (8 of 17)** on test, and 8 of the 17 were called Humour. The overall 73.3% has a margin of about +/- 11 points with 60 items, so it is not clearly better than the dev score. The earlier v1 run cannot be reproduced: its prompt text was not saved, and the file now in the repo uses a much longer prompt than that run did.

6. Cost model

Cost per comment = (prompt tokens x input price + completion tokens x output price), measured from the tokens of each run at Groq's list price for the model, **\$0.075 per 1M input tokens and \$0.30 per 1M output tokens** (checked October 2026; prices change).

Run	Tokens per comment (prompt + answer)	Cost per 1k comments	At 100x (100k comments)	Median latency
v1 (earlier run)	227 + 232	\$0.09	\$8.67	0.8 s
v2	692 + 413	\$0.18	\$17.58	3.0 s
v3 (dev)	1,009 + 514	\$0.23	\$22.99	4.0 s
v3 (test)	1,010 + 378	\$0.19	\$18.91	1.0 s

v3 costs about 31% more per comment than v2 and scores 15.7 points higher on dev. At 100x volume (100,000 comments) the classifier costs about \$19 to \$23. The judge used 40,964 tokens for its 40 grading calls (under \$0.02), about 1,000 tokens per judged answer, so judging every answer would add roughly two thirds to the cost of v3. The free Groq plan limits tokens per minute (8,000) and per day (200,000), which allowed about one full dev run per day; latency figures include retry waits after rate-limit errors, so we report the median.